

Confidence intervals, t-score

Background You are given the dataset from the lesson

Task 1 Calculate the mean and the standard error of the dataset

Task 2 Determine which statistic to use for inference

Task 3 Find the appropriate statistic, taking into consideration the degrees of freedom (if applicable) for 99% confidence

Task 4 Find the 99% confidence interval

Solution:

Dataset	
\$	78,000
\$	90,000
\$	75,000
\$	117,000
\$	105,000
\$	96,000
\$	89,500
\$	102,300
\$	80,000

Task 1: Mean \$ 92,533

St. deviation \$ 13,932

Standard error \$ 4,644

Task 2: Population variance is unknown

We have a small sample

We assume that the population is normally distributed

The appropriate statistic to use is the t-statistic

Task 3: 99% CI, $t_{8,0.005}$ 3.355

Sample size is 9 => the degrees of freedom are 8.

Task 4:

	T	CI low	CI high
99%	\$	76,953	\$ 108,114